FVS-ACD v12.11: the maturity-dependent bias, addressed

Five gated levers, byte-identical gate-off, all four guardrails pass

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Honest read: v12.11 wins on managed and mature stands, beating OSM v2.26.1 where it counts. On developing Maine FIA, OSM still leads, and finer subplot grain does not close the gap. v13.1 targets it next.

CONTEXT

v12.10.1 ships; v12.11 addresses the bias that flips sign with maturity

2018 (Hennigar & Weiskittel, ECANUSA). OSM trailed only slightly behind FVS-ACD, and beat it on merchantable volume-increment bias and RMSE in ME, NB, and NS. That result set the FVS-ACD to-do list.

2026. v12.10.1 ships as the production model. Its successor, v12.11, targets the one bias that survived every prior fix: a long-horizon volume error that flips sign across the stand-maturity gradient.

1. **Site productivity** ESI holds; height edge is extrapolation
2. **Data harmonization** MagPlot + FIA on one NSVB currency
3. **New calibration approaches** Five gated levers correct the maturity-dependent bias
4. **Integrating climate** Current and 2030 ESI as a scenario
5. **AI and productivity** Accelerated the work, caught our own error

His five agenda topics, and what we bring to each

DATA HARMONIZATION

Data harmonized onto one currency

705

ESI-covered in-domain stands, the pooled NSVB scorecard sample

NSVB

FIA volume and biomass as the common currency for every model

ESI v8

30 m climate-and-terrain site driver, cross-border coverage

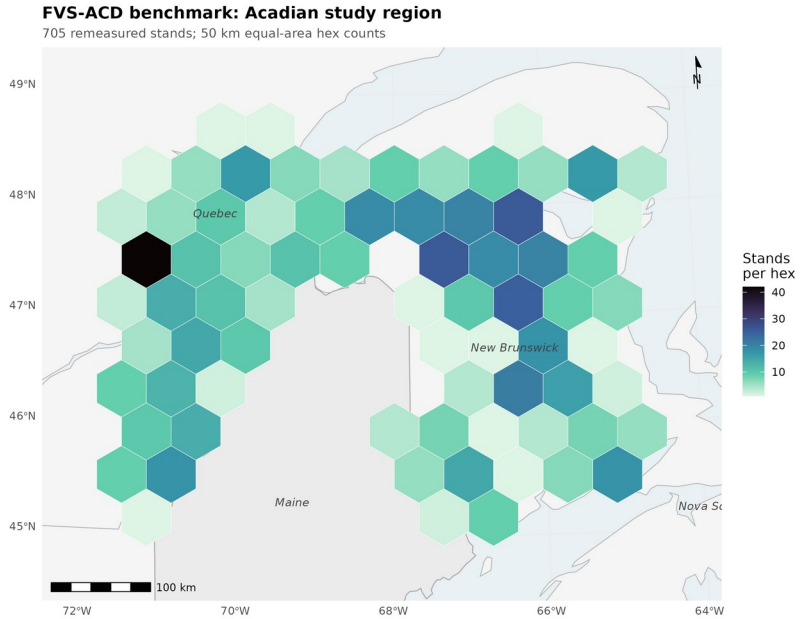
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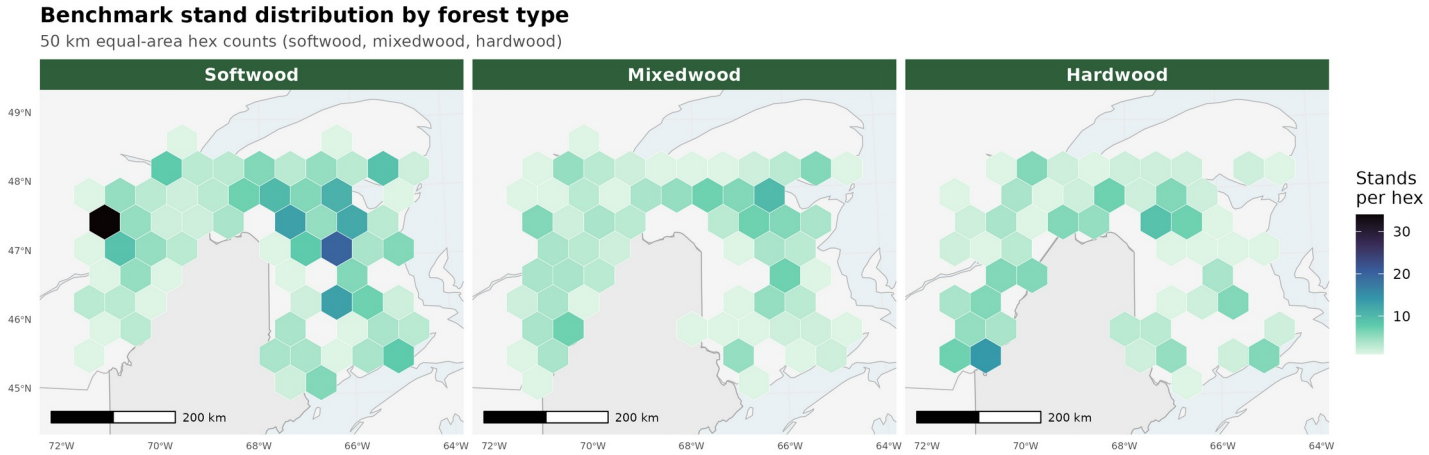
models scored head to head v12.5.0, v12.7.0, v12.9.x, v12.10.1, OSM

Also: recent post-2010 Maine FIA (independent in time from the 1999 to 2005 training FIA), and SILC Seven Islands for the long horizon. Coordinates stay on Cardinal; aggregates only.

BENCHMARK DATASET

One harmonized benchmark: 705 remeasured Acadian stands





705
remeasured stands

355 / 350
New Brunswick / Quebec

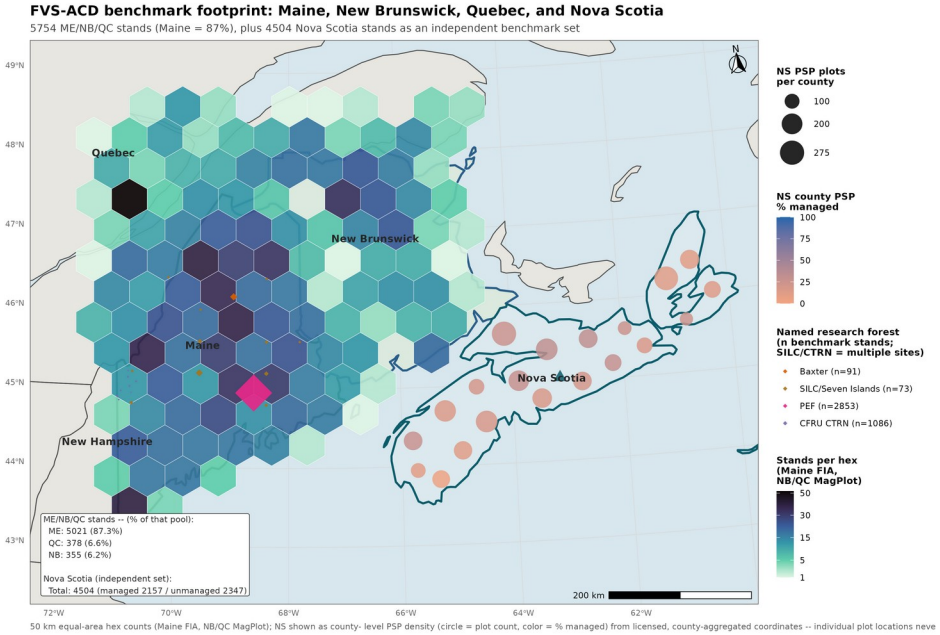
321 / 186 / 178
softwood / mixedwood / hardwood

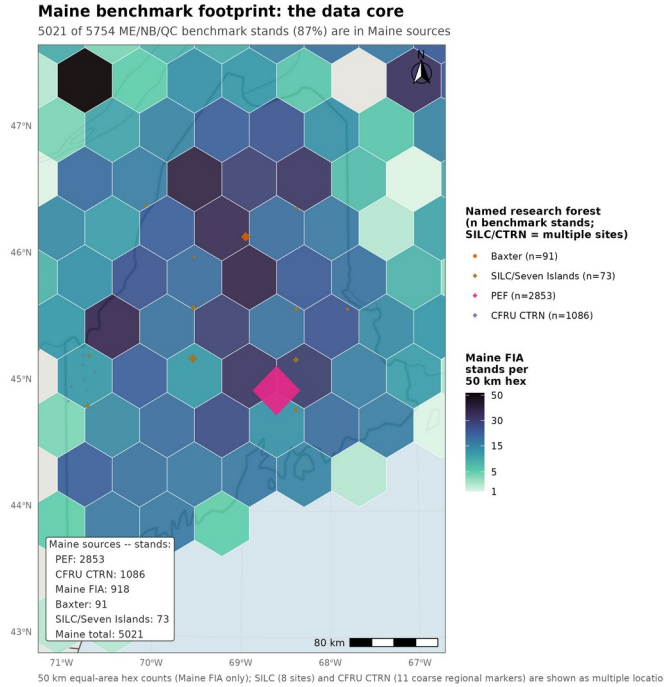
4 to 14 yr
remeasurement interval

Coordinates never released: locations are aggregated to a 50 km equal-area hex grid; true plot coordinates stay on the analysis cluster.

DATA FOOTPRINT

The v12.11 benchmark is Maine heavy, 87.3%, plus an independent Nova Scotia test





5,021 / 5,754
Maine stands, 87.3% of the pooled ME/NB/QC total

355
New Brunswick stands

378
Quebec stands

4,504
Nova Scotia stands, managed and unmanaged (independent test)

8
sources across four jurisdictions

The v12.11 candidate benchmark pools seven sources across Maine, New Brunswick, and Quebec into one three-way scored comparison, base, candidate, and OSM v2.26.1. Nova Scotia is evaluated separately as a fully independent, fourth-jurisdiction test (Nova Scotia section, later in this deck), never used to fit or tune any lever. Coordinates never released: true plot locations stay on the analysis cluster, aggregates only; Nova Scotia's PSP locations are additionally restricted under its own provincial data-license agreement.

DATA FOOTPRINT

Within Maine, PEF and CTRN anchor a five source core

Stand counts by source

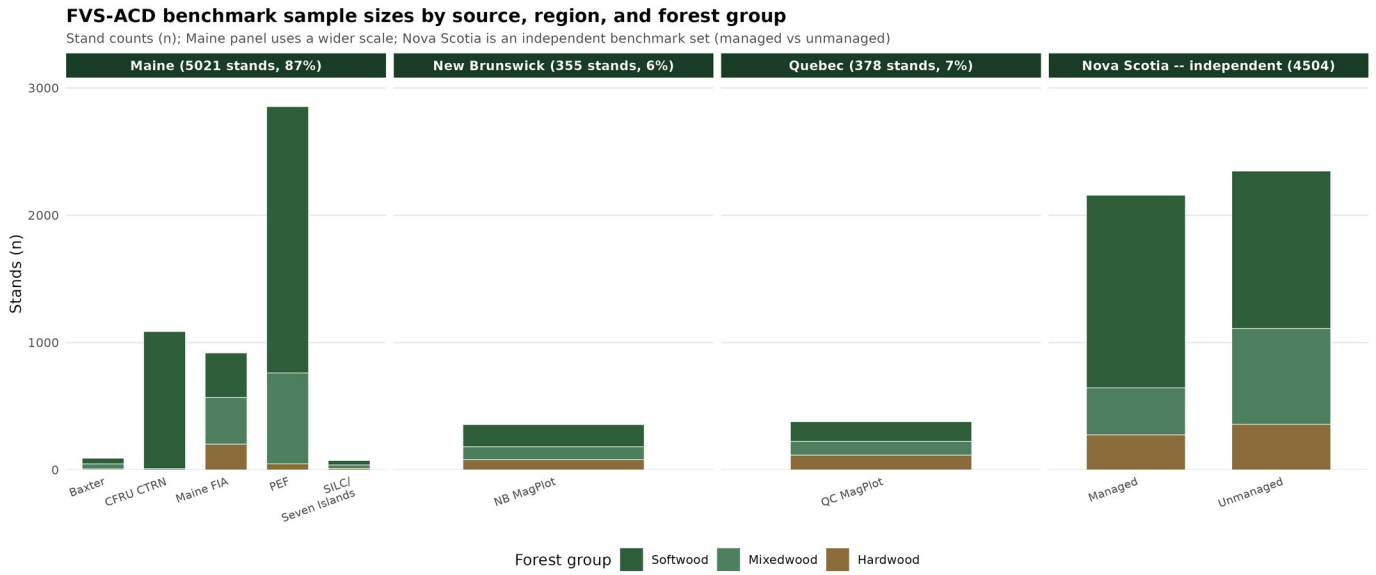


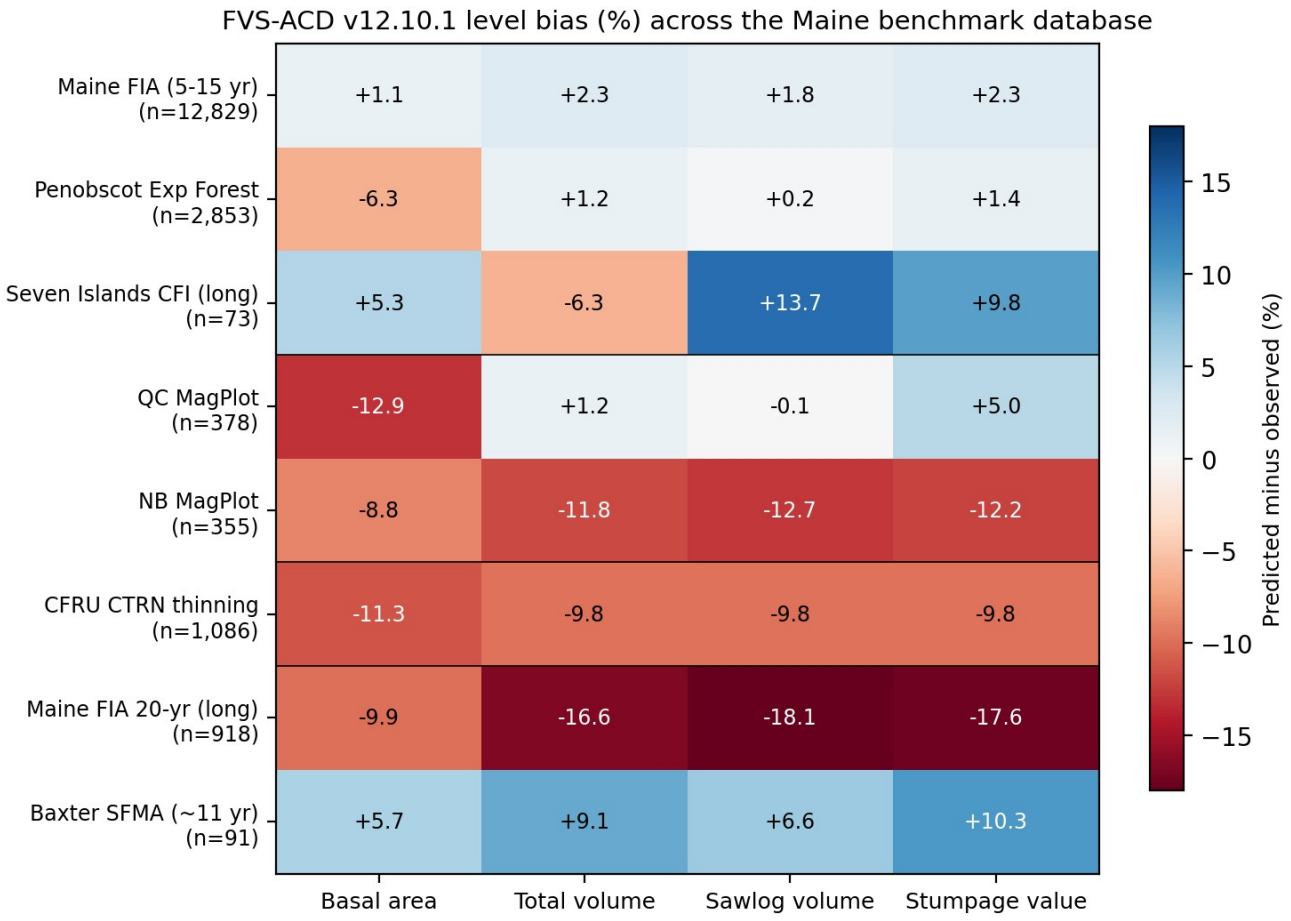
Figure. Stand counts by source, the seven ME/NB/QC sources scored in the forest-group and overall synthesis slides, plus Nova Scotia's independent managed/unmanaged split (4,504 stands, not pooled into the ME/NB/QC total).

Maine's 5,021 stands split five ways: Penobscot Experimental Forest 2,853, CFRU CTRN 1,086, Maine FIA 918 at whole plot grain, Baxter State Park 91, and SILC Seven Islands 73. New Brunswick MagPlot adds 355 and Quebec MagPlot adds 378, completing the seven source, 5,754 stand pooled benchmark. Nova Scotia adds an eighth source and fourth jurisdiction, scored independently: 4,504 stands, 2,157 managed and 2,347 unmanaged.

CTRN carries no hardwood classified stands in this benchmark, 1,077 softwood and 9 mixedwood only. Maine FIA is also re-scored at a finer 2,858 stand subplot grain to test whether the developing end fix survives finer aggregation, a separate population from the 918 whole plot stands counted here.

MULTI-SOURCE BENCHMARK

FVS-ACD validates across a multi-source Maine benchmark database



Best on independent Maine data

Short-interval total-volume bias runs +2.3% on Maine FIA and +3.7% at PEF, better than the Quebec and New Brunswick pairs the model was tuned against (NB -11.8%, CTRN -9.8%).

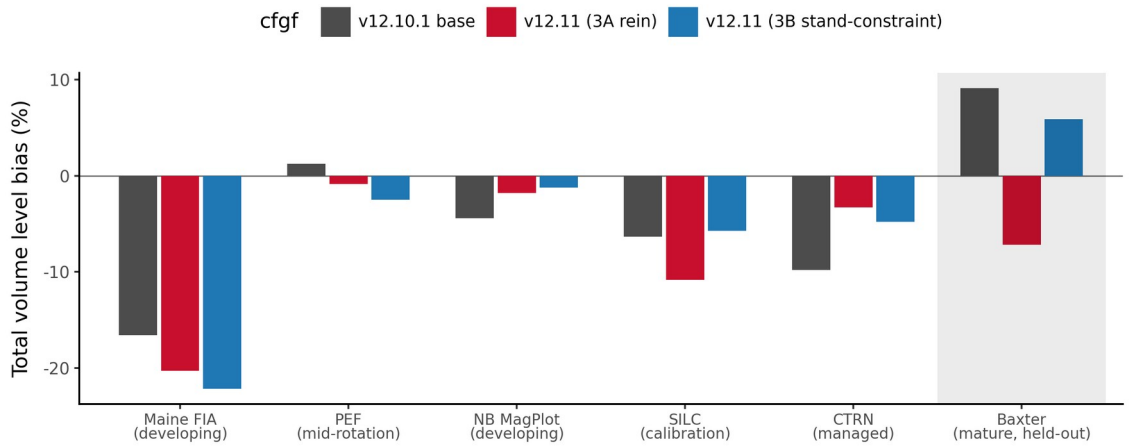
Eight sources, one NSVB currency

Maine FIA (n=12,829 short, 918 at 20 yr), PEF (n=2,853), Seven Islands CFI long horizon (n=73), Baxter SFMA reserve (n=91), plus Quebec and New Brunswick MagPlot and CFRU CTRN thinning.

THE PROBLEM

v12.11 tames the mature-stand over-prediction

Total volume bias



Basal area bias

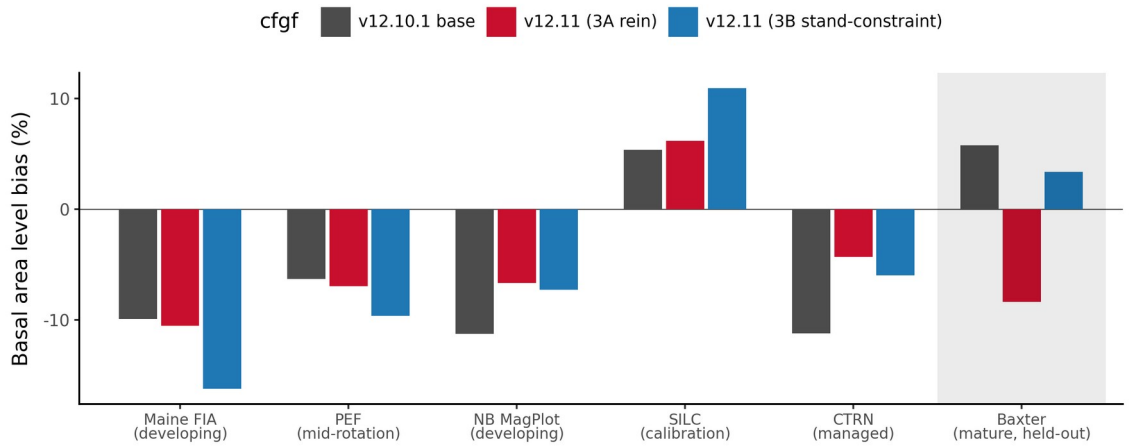


Figure 1. Total volume and basal area level bias across the stand-maturity gradient, base v12.10.1 versus the v12.11 candidate.

In the shipped v12.10.1 baseline, total-volume bias moves from -16.60% on developing Maine FIA stands (whole-plot grain), through -9.78% on managed CTRN, to +9.12% on the held-out mature Baxter reserve, without v12.11 reopening that sign flip: it narrows the mature-end over-prediction (Baxter +9.12% to +6.92%) and roughly halves the managed-end bias (CTRN -9.78% to -2.24%), the sign flip the next slide's decomposition traces to three separable component errors. The developing end barely moves (Maine FIA -16.60% to -16.13%) and remains the acknowledged gap v13.1 targets next.

Volume bias moves from -16.7% on the Maine FIA 20 year set and -8.3% at PEF long, through -3.4% at Seven Islands CFI, to +9.1% at the mature Baxter reserve. FVS-ACD under-predicts in developing, vigorous stands and over-predicts in mature, senescing reserves, but a component decomposition shows this flip is the net of three separable errors: survivor diameter growth runs too high, an ingrowth deficit dominates the developing-stand under-prediction, and the SILC-calibrated mortality over-removes at managed and high-density structures. At the mature Baxter end the over-prediction is survivors over-accreting girth on large old trees, with mortality essentially exact there, not too little removed (next slide). OSM swings the same way (PEF -4.5%, SILC -14.9%, Baxter +12.4%), a shared growth and yield problem and exactly what the v13.1 mortality and ingrowth refit targets. It strengthens the case for the mature Acadian remeasurement data, the August ask.

DECOMPOSITION

Three separable errors, survivor growth, recruitment, mortality, map to three levers

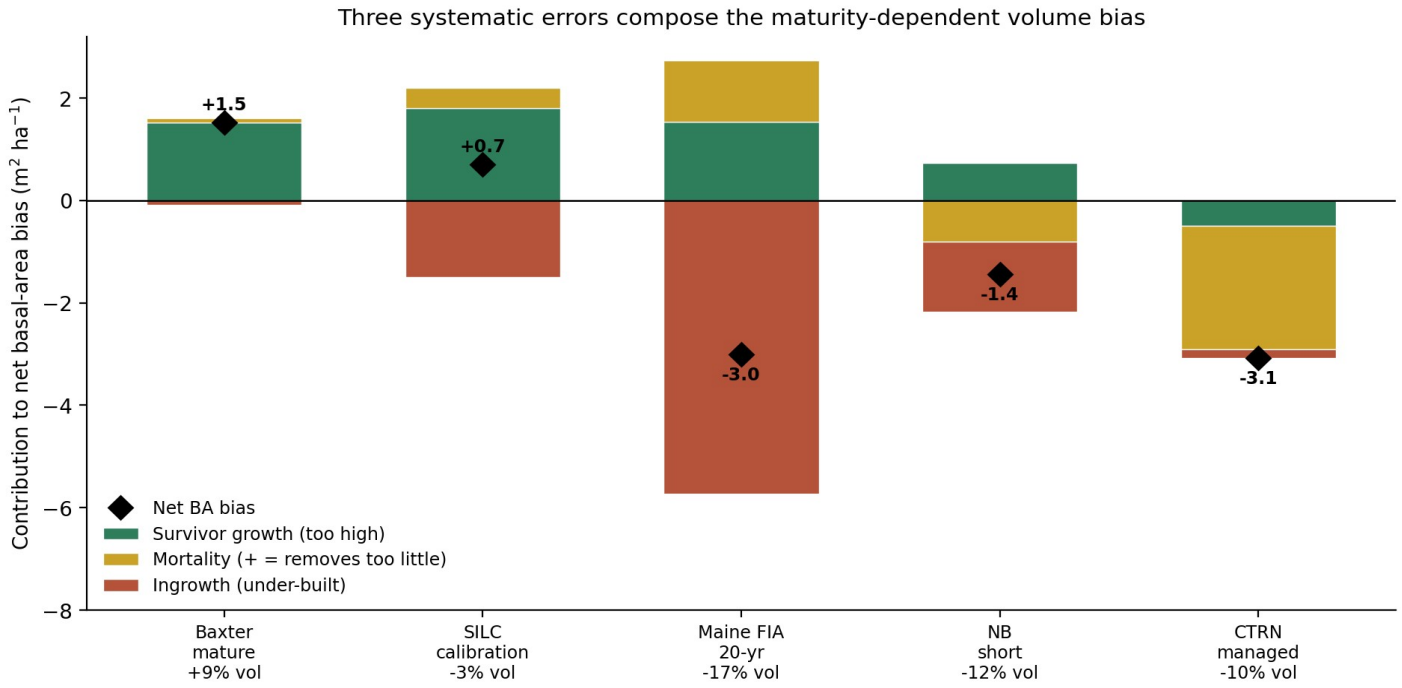


Figure 2. Component decomposition of net stand volume change, observed versus predicted, base versus the v12.11 candidate, for Maine FIA, Baxter, and CTRN.

Survivor growth too high -> Lever 3

Diameter increment over-accretes girth on large old trees on every source; at mature Baxter mortality is essentially exact. The Chapter 10 QMD-gated stand constraint reins it in only where QMD is high enough to matter.

Recruitment deficit -> Lever 1

Predicted recruitment 7.6 vs observed 38.5 m^3 ha^{-1} on the Maine FIA 20-yr set drives the developing-end under-prediction. Lever 1 lifts the flat, basal-area-anchored ingrowth deficit (about 2.5x pooled, IGMULT = 2.5).

Mortality over-removal -> Lever 2

SILC-calibrated self-thinning over-removes at managed, high-density structures: 4.6x observed at CTRN, 2.5x at NB. Lever 2 attenuates the removal fraction only as relative density climbs past onset (RD0 = 0.38); natural stands are untouched.

Seven Islands SILC, the calibration set, nets near zero by cancellation of the survivor-growth overshoot and the ingrowth deficit. The QMD floor (SC_QMIN = 22) was added after this decomposition showed an un-gated stand constraint suppressed developing-stand growth.

THE FIX

Five gated levers, each off by default, on only where the biology calls for it

1	**Recruitment**	Flat, basal-area-anchored ingrowth deficit pooled about 2.5x (IGMULT = 2.5). A gated RD taper protects mature closed stands but is not enabled in the scored configuration.
2	**Management-gated mortality**	Attenuates the self-thinning removal fraction as relative density climbs past onset: f(RD) = fmin + (1 - fmin) x exp(-k x max(RD - RD0, 0)); fmin = 0.13, k = 12.4, RD0 = 0.38. Natural stands are untouched.
3	**Chapter 10 QMD-gated stand constraint**	Lowers the activation threshold on the proportional-increment constraint to a stand basal-area-increment target, gated by a 22 cm QMD floor so it binds only in mature, high-QMD stands and spares developing stands.
4	**Hardwood height by form and risk**	High-risk trees run about 10 to 11% shorter in *Betula alleghaniensis* Britton and *Acer saccharum* Marshall (both $p < 0.002$); weaker, not significant in *Acer rubrum* L. ($p = 0.59$). Fitted from 635 field trees across four Maine installations.
5	**Crown ratio, investigated**	Already active: crown ratio is projected dynamically every cycle by the Russell, Weiskittel and Kershaw (2014) recession model and feeds diameter and height increment. The changelog note describing it as disabled was four years stale, not a new lever.

ENGINEERING RIGOR

Every lever is gated off by default, and gate-off reproduces v12.10.1 bit for bit

0.000e+00

maximum absolute difference,
candidate vs frozen v12.10.1 baseline

Confirmed on a 20-year projection of both a young and a mature stand after every engine change, and reconfirmed after the red-team fixes on a novel hardwood-heavy stand not used in any prior smoke test: *Betula alleghaniensis* Britton, *Acer saccharum* Marshall, *Acer rubrum* L. mix, CSI 20, 30-year projection, on diameter, height, and expansion factor.

VALIDATION

All four guardrails pass, with two flags disclosed, not hidden

Guardrail	Base v12.10.1	v12.11 candidate
1. Frozen in-domain FIA NSVB (n=733)	BA -11.28%, TOPHT +4.21%, TOTVOL -4.41%	BA -7.31%, TOPHT +5.88%, TOTVOL -1.50%
2. Bakuzis five relations (Reineke slope)	-1.59, PASS	-1.62 primary PASS; -1.88 managed sensitivity, FLAG only
3. Long-horizon plateau (~160 yr)	BA 36.10 $m^2\ ha^{-1}$, plateaus	BA 35.29 $m^2\ ha^{-1}$, plateaus
4. Managed out-of-sample vs OSM v2.26.1	Trails OSM on all 4 cells	Beats or ties OSM on 3 of 4 cells

PASS on every guardrail. The TOPHT drift traces to the always-on per-province dht.mult height-increment scalar (NB_AFAC=0.579, QC_AFAC=0.111), not to the new growth levers and not to an ESI-consistent anchor-recalibration mechanism, which does not exist in this model file (next slide, red-team reframe). The lone Reineke flag is confined to the managed sensitivity check, just beyond the -1.8 to -1.4 band, and does not block the primary configuration. On the managed cell v12.11 ties OSM on bias but still trails on RMSE, though the gap narrowed from a 9.4-point deficit under the base to 3.5 points.

CENTRAL RESULT

On managed and mature stands, v12.11 beats OSM v2.26.1

Source	n	Total volume bias base / v12.11 / OSM	Basal area bias base / v12.11 / OSM
Baxter, ME (mature, held out)	91	+9.12% / +6.92% / +12.37%	+5.74% / +4.47% / +13.28%
CTRN, ME (managed)	1,086 / 1,033	-9.78% / -2.24% / -4.81%	-11.26% / -3.52% / -4.19%
NB MagPlot (managed)	355	-11.84% / -5.96% / -8.07%	-8.84% / -2.33% / -2.48%

Two managed flags, disclosed not hidden: Quebec MagPlot basal area improves sharply (-12.95% to -4.12%) but total volume worsens to +9.46%, the management gate over-corrects volume relative to basal area there; SILC Seven Islands basal area widens (+5.35% to +11.02%). Both are named v13.1 targets, not concealed.

Baxter and CTRN grain: stand. NB MagPlot grain: stand. QC MagPlot has no scored OSM comparison this round.

CENTRAL RESULT

OSM still leads developing Maine FIA; finer subplot grain does not close it

Source	Grain	n	Total volume bias: base / v12.11 / OSM
PEF, ME (developing)	stand	2,853	+1.23% / +1.37% / +0.26%
Maine FIA (developing)	whole-plot	918	-16.60% / -16.13% / -6.29%
Maine FIA (developing)	subplot (~0.017 ha)	2,858	-24.90% / -24.82% / no OSM comparison

OSM leads the developing whole-plot comparison by a wide margin (-6.29% versus the candidate's -16.13%), a gap the recruitment lever narrows only modestly from the base's -16.60%. At the finer FIA subplot grain the gap barely moves at all, -24.90% to -24.82%, a 0.08 percentage point change, against the whole-plot improvement of about 0.5 points. The developing end, whole-plot and especially subplot, is the acknowledged open gap in this candidate, and the explicit target of v13.1, not a claim closed by v12.11.

THE HONEST HEADLINE

v12.11 wins where it was built to win, and says plainly where it does not

- v12.11 wins clearly on the managed sources: CTRN and NB MagPlot see bias roughly halved on both basal area and total volume against the frozen base, with lower RMSE, and now beat OSM v2.26.1 on both sites rather than trailing it.

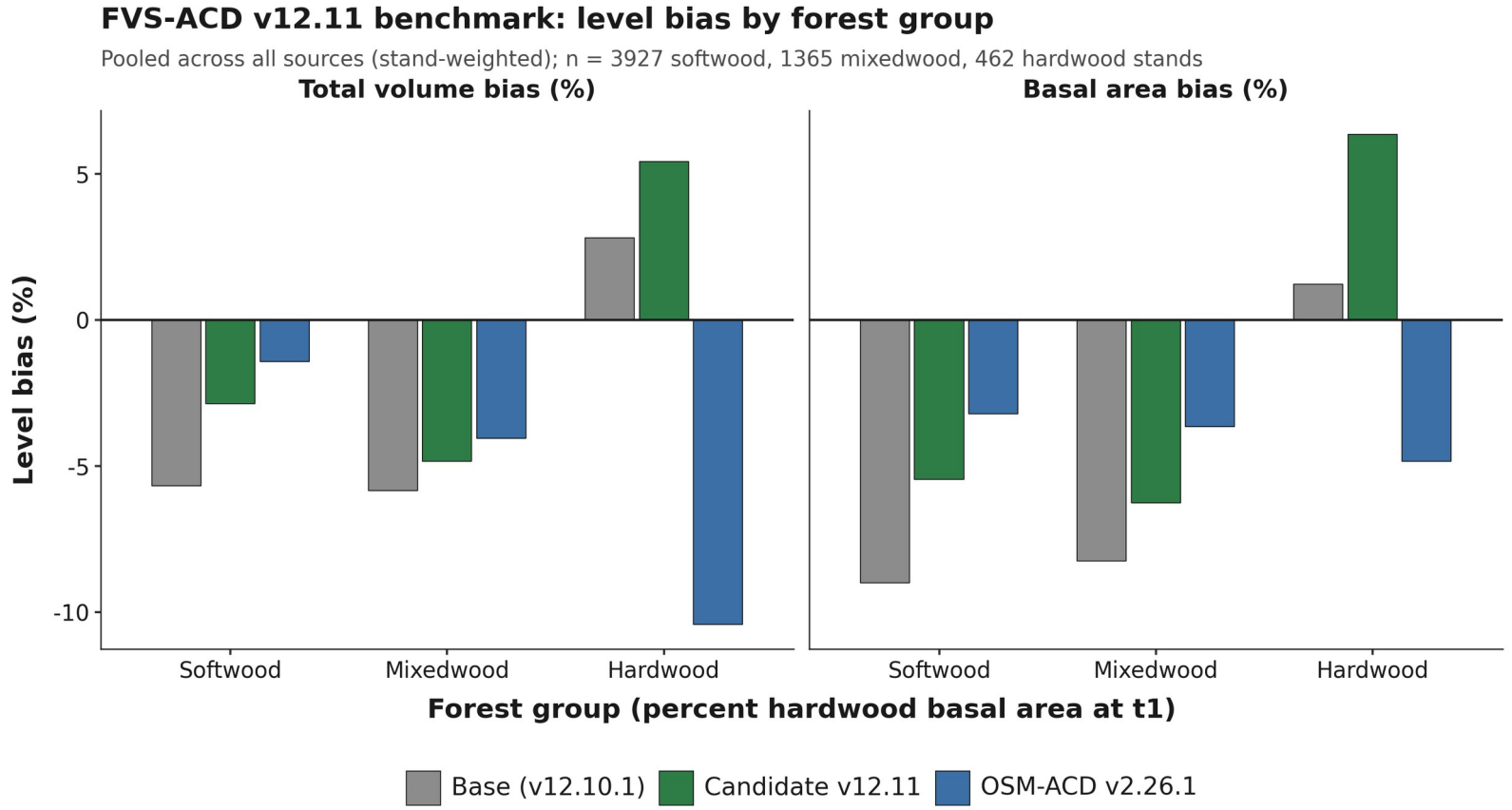
- The held-out mature Baxter installation, never used in fitting, moves toward zero (total volume +9.12% to +6.92%, basal area +5.74% to +4.47%) and beats OSM by a wide margin (OSM +12.37% volume, +13.28% basal area).

- Two managed flags, disclosed not hidden: Quebec MagPlot's basal area improves sharply but its volume bias worsens to +9.46%, an over-correction; SILC's basal area widens. Both are named v13.1 targets.

- Where v12.11 does not win: OSM remains ahead on developing Maine FIA at whole-plot grain, and at the subplot grain the gap barely moves at all. The developing end is the acknowledged open gap, and the explicit target of v13.1, not a claim closed by v12.11.

FOREST-GROUP PERFORMANCE

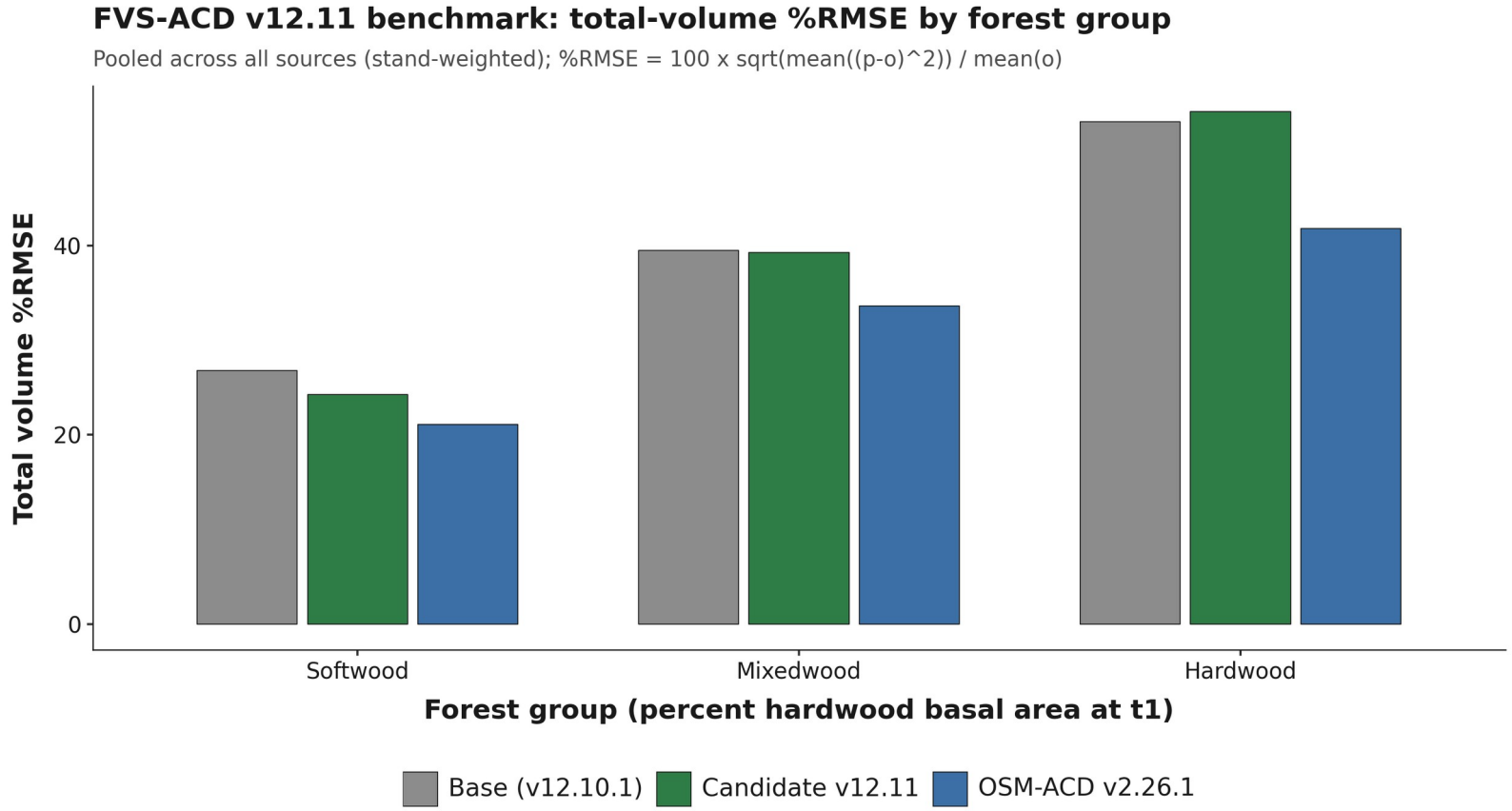
Softwood is v12.11's clean win; the pooled hardwood shift is mostly a QC-source artifact



Pooled across all seven sources, softwood total-volume bias improves from -5.67% to -2.86% and basal-area bias from -9.00% to -5.45%, the narrowest gap to OSM (-1.43%, -3.21%) of the three forest groups. Mixedwood gains a smaller but real margin. Hardwood's pooled bias does widen (+2.82% to +5.42% total volume), but QC MagPlot alone drives roughly three-quarters of that shift, and QC's candidate-vs-base move is uniform across all three forest groups (+7.6 to +8.8 points in softwood, mixedwood, and hardwood alike), a pattern the hardwood-only Lever 4 cannot produce. With QC set aside, the true cross-source hardwood signal is small and mixed (base -0.77% to candidate +1.28%, n=347: Baxter, Maine FIA, PEF, SILC, New Brunswick), with three of five non-QC sources improving. n = 3,927 softwood, 1,365 mixedwood, 462 hardwood.

FOREST-GROUP PERFORMANCE

Precision tightens for softwood and mixedwood; hardwood is an open question, not a settled regression



Percent RMSE falls with the bias gains on softwood, 26.79% to 24.25%, and mixedwood, 39.46% to 39.23%, though OSM keeps the tighter scatter on both, 21.08% and 33.61%. Hardwood ticks the wrong way on precision too, 53.06% to 54.12%, while OSM's precision is closer, 41.77%, even as its bias swings negative. This should not be read as a confirmed hardwood-specific v13.1 target: the bias-side decomposition shows the pooled hardwood shift is substantially a single-source (QC) artifact, not a genuine cross-source hardwood degradation, and the same province-assembly check has not yet been run against the precision numbers. Softwood remains this benchmark's clear win.

OVERALL ACROSS ALL DATASETS

Pooled across every dataset, v12.11 nearly closes the bias gap to OSM; precision does not follow

Source	n candidate / OSM	Total volume bias base / v12.11 / OSM	Basal area bias base / v12.11 / OSM
Softwood	3,927 / 3,716	-5.67% / -2.86% / -1.43%	-9.00% / -5.45% / -3.21%
Mixedwood	1,365 / 1,257	-5.84% / -4.84% / -4.05%	-8.25% / -6.26% / -3.66%
Hardwood	462 / 346	+2.82% / +5.42% / -10.42%	+1.23% / +6.35% / -4.83%
All forest types (computed)	5,754 / 5,319	-5.03% / -2.66% / -2.63%	-8.00% / -4.70% / -3.42%

The all forest types row is a stand-weighted mean of the three rows above, computed from the forest group scorecard, not a separately scored pooled run. On that basis v12.11's overall bias, -2.66% total volume, is now close to OSM's -2.63%, but percent RMSE tells a different story: v12.11 pools to 30.21% against OSM's 25.39%. The hardwood row driving part of that RMSE gap is substantially a QC MagPlot artifact rather than a genuine hardwood-specific effect (previous slide); softwood and mixedwood carry the pooled row's real, source-consistent gains.

Base = frozen v12.10.1. Candidate = v12.11. OSM = OSM-ACD v2.26.1. Source: figures/_data/v1211_forestgroup_scorecard.csv, pooled stand-weighted rows by forest group.

RED TEAM

Three of six reframes with numeric or methods impact

Top height is an anchor, not a lever

Six lever configurations run through an identical stand gave bit-identical top height to three decimals (20.269 m) regardless of which levers were on. The ESI dominant-height anchor forces top height onto a curve driven by climate site index and age alone. A second-round check traced the drift to the always-on per-province dht.mult height-increment scalar (NB_AFAC=0.579, QC_AFAC=0.111), not to an ESI-consistent anchor-recalibration mechanism, which does not exist in this model file despite the harness still logging ANCHOR=TRUE.

Lever 4 is a species average, not tree-specific

Every reported Lever 4 number was scored against SPLIT-synthesized form and risk: an identical population-average, expansion-factor-weighted multiplier applied to every stem of a species regardless of its individual size, vigor, or condition. Describe it as a species-average height correction, not tree-level risk conditioning.

Quercus rubra's coefficient was capped

Its worst-case pre-cap multiplier was a 26.7% height derate, more than double the two species with genuine field validation (about 12.6 to 12.7% for *Betula alleghaniensis* Britton and *Acer saccharum* Marshall), and rested entirely on an FIA damage-code proxy because *Quercus rubra* L. had only one tree in the Smith field set, dropped. Capped to the SM/YB field magnitude pending independent field heights.

Independent verification reconfirmed byte-identical gate-off on a novel stand after every fix (max|Δdbh| = max|Δht| = max|Δexpf| = 0.000e+00). A fourth reframe corrects the hardwood forest-group causal story to a QC-source artifact (next section). A fifth confirms byte-identical gate-off a second time after a gated ESI-anchor height-cap re-clamp; it changes no number reported here. A sixth catches Nova Scotia's apparent long-interval over-prediction as a contamination artifact; the clean record shows attenuation, not a sign flip (Section 11).

Percent bias inflates on low means; %RMSE is the fairer read

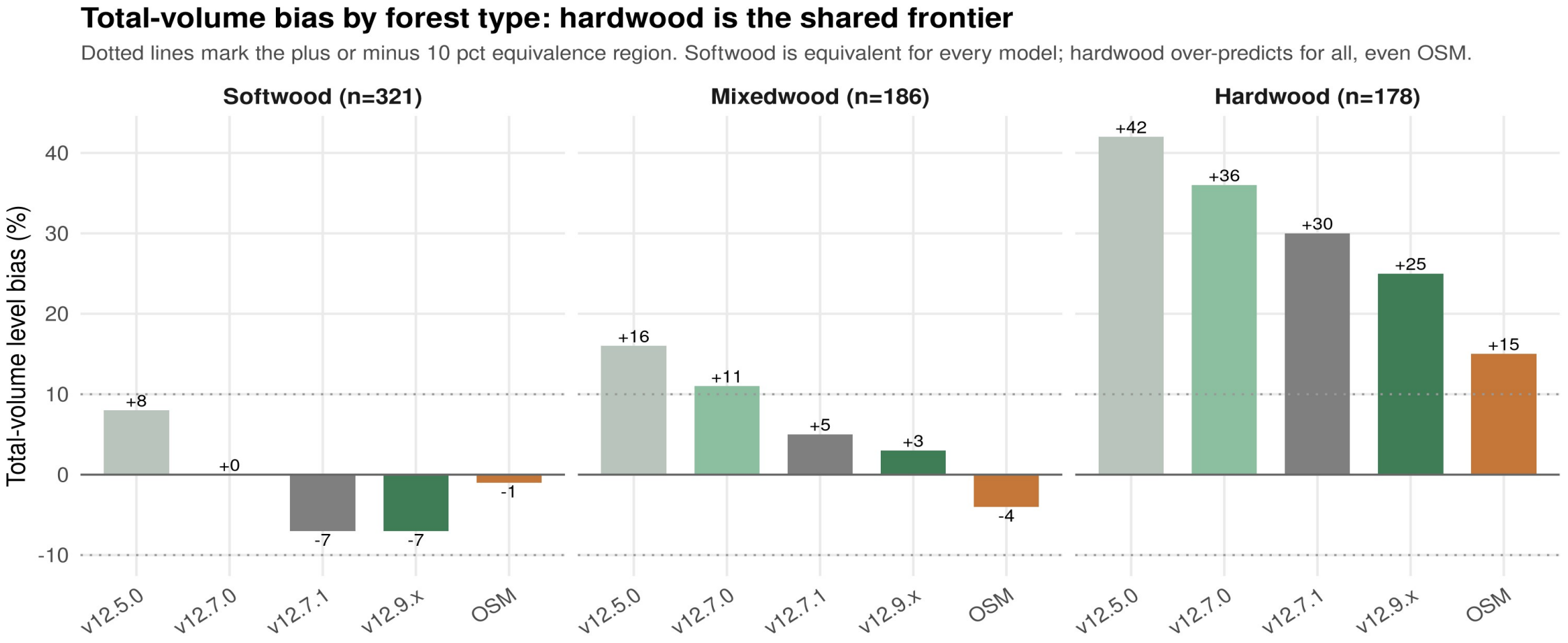
Figure 11. Percent RMSE scorecard (%RMSE = 100 x RMSE / mean observed), nine NSVB metrics by model on the operational in-domain set (n = 705). Lower is better; brackets are within-province bootstrap 95% intervals. %RMSE blends accuracy and precision, so it does not deflate on high-mean metrics.



Percent bias divides by the mean, so it balloons on low-mean metrics and can read a model as far worse than it is. Percent RMSE blends accuracy and precision and does not deflate. Hardwood is a high-variance frontier shared by every model: v12.10.1 hardwood total volume is +19 percent bias but about 76 percent %RMSE, and OSM is about 75 percent.

LEGACY VALIDATION, PRE-V12.11

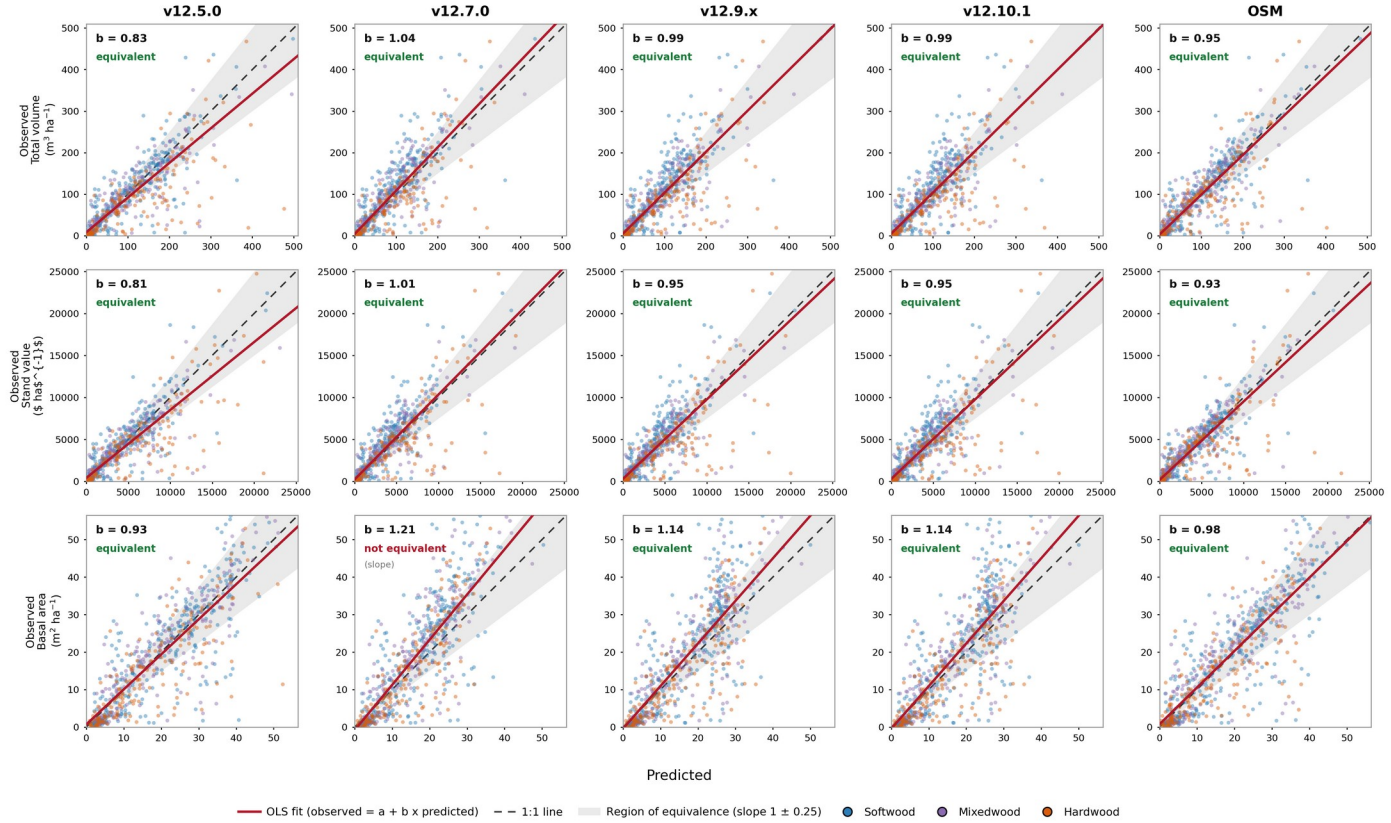
Hardwood was already the shared frontier before v12.11



Softwood is equivalent for every model. Hardwood is the shared frontier: all models over-predict, even OSM (+15). v12.9.x roughly halves the v12.5.0 hardwood bias. This is the legacy in-domain 705-stand benchmark scored across v12.5.0 through v12.9.x, predating v12.11; the forest-group slides earlier in this deck quantify the same hardwood frontier for the current candidate against base and OSM.

Four of five models pass all nine equivalence metrics

Figure 10. Observed versus predicted with the Robinson and Froese region of equivalence, FVS-ACD frozen benchmark (POOLED in-domain, n = 705). The joint verdict requires both the slope (1 ± 0.25) and the intercept (within ±25% of mean observed) confidence intervals to fall inside their equivalence regions.

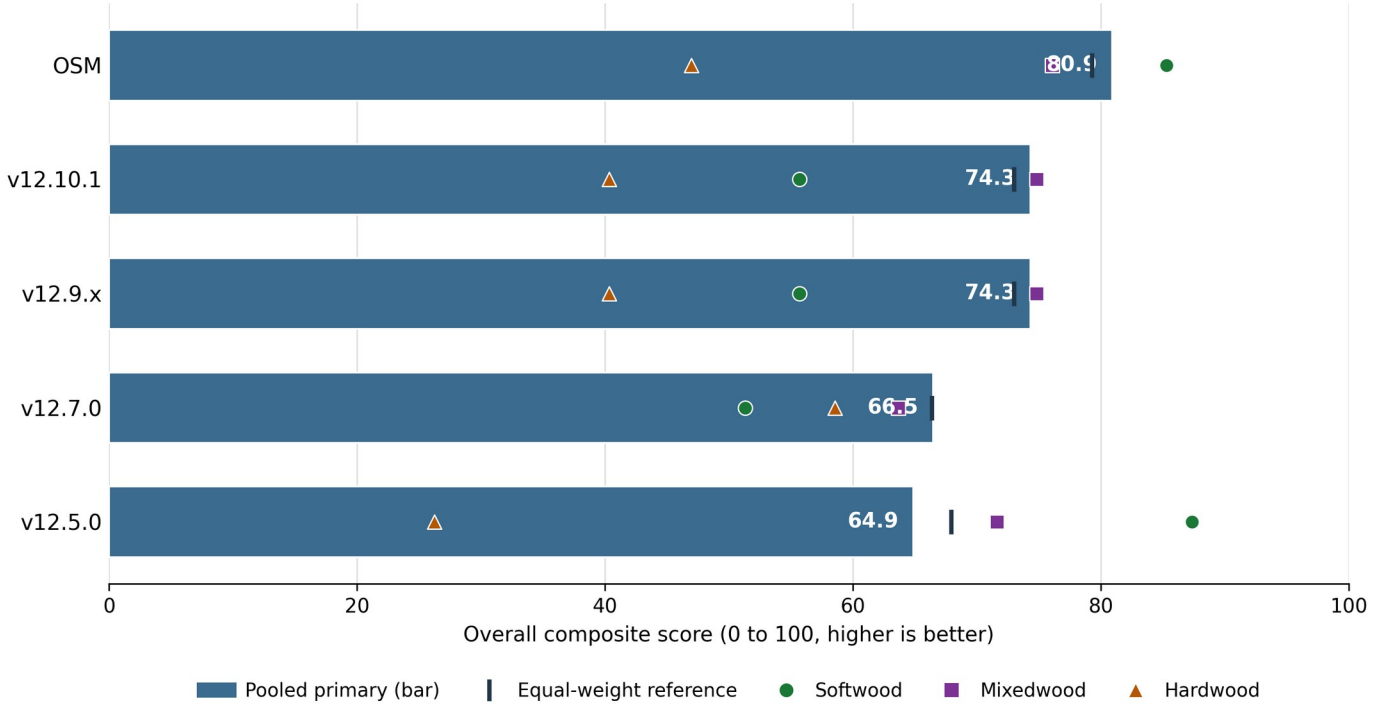


Observed on predicted with the Robinson and Froese region of equivalence, joint on slope (1 ± 0.25) and intercept (within 25 percent of the mean observed). Four of five models pass all nine metrics. The one exception is v12.7.0 basal area, where the slope runs steep at 1.21. Every volume and value metric passes for every model. This equivalence battery has not yet been re-run for v12.11; it remains the methodology this deck uses to judge slope and intercept equivalence, applied here to the model lineage through v12.10.1 and OSM.

OVERALL COMPOSITE, LEGACY 705-STAND BENCHMARK

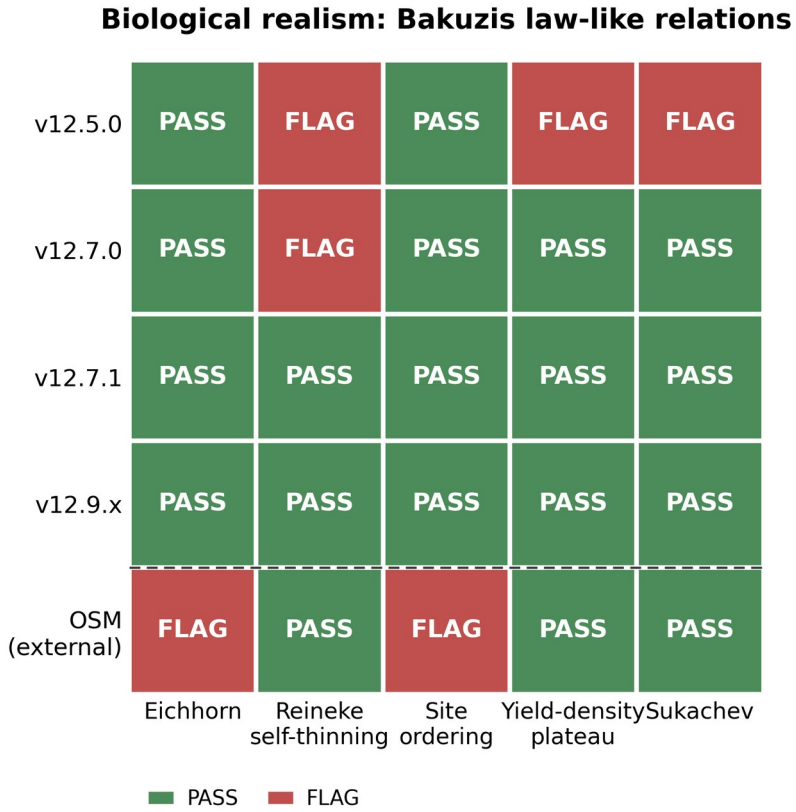
OSM leads either composite; the FVS order flips by view

Figure 8. Volume and value weighted overall composite score by model: pooled in-domain primary composite (bars, n = 705), equal-weight reference (ticks), and by-forest-type primary composites (markers)



Volume and value carry the most weight. On the %RMSE-weighted composite, which blends accuracy and precision, the order is OSM 53.4, v12.7.0 52.1, v12.9.x and v12.10.1 tied at 49.8, v12.5.0 49.4. On the bias-weighted view shown above it is OSM 80.9, the shared v12.9.x and v12.10.1 build 74.3, v12.7.0 66.5, v12.5.0 64.9. v12.10.1 is the least biased; v12.7.0 has the tighter scatter. This composite scores the model lineage through v12.10.1 against OSM on the narrower in-domain 705-stand set and predates v12.11; the pooled forest-group synthesis for the current candidate, on the full 5,754-stand all-sites benchmark, appears earlier in this deck.

Biological realism: only the self-thinning versions pass all five



Green passes, red flags. OSM (external) is scored on all five relations from its own 150-plus-year run over the SILC/CFI stands binned into three T50HT site terciles; it passes self-thinning, the yield-density plateau, and Sukachev, and flags Eichhorn and site ordering under the real-stand substitution (projection-year axis, heterogeneous starting ages).

The FVS-ACD self-thinning versions plateau at the realistic maximum and pass all five relations in the seeded-gradient battery. OSM, run 160 years, is now fully scored on all five: it passes Reineke self-thinning, the yield-density plateau, and Sukachev, and flags Eichhorn and site ordering.

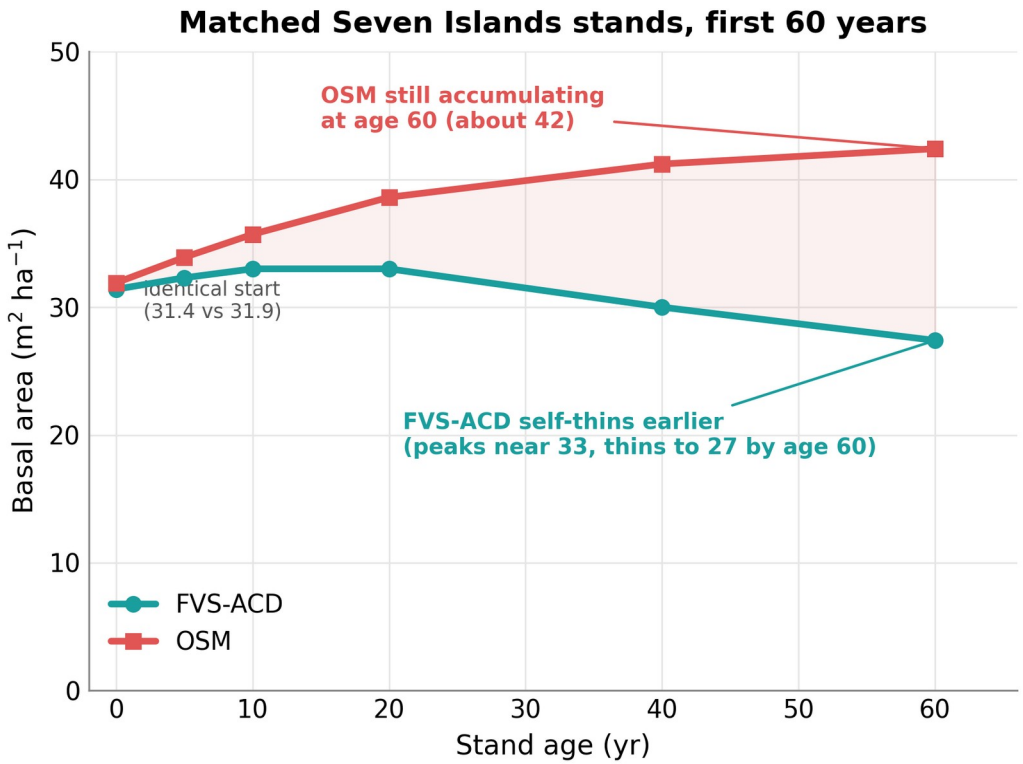
v12.5.0 never self-thins (basal area runs to 65 with thousands of stems alive), failing yield-density, Reineke, and Sukachev. The age-60 read that showed OSM in runaway accumulation caught only the pre-plateau limb; over the full horizon OSM plateaus and self-thins.

Honest limit: ingrowth. Recruitment is under-predicted (NB 17.9 vs 39.8, QC 11.7 vs 19.7 stems ha^{-1} yr^{-1}). Direction is correct, it responds to density and management; the level is low. A disclosed limit and a v13 target.

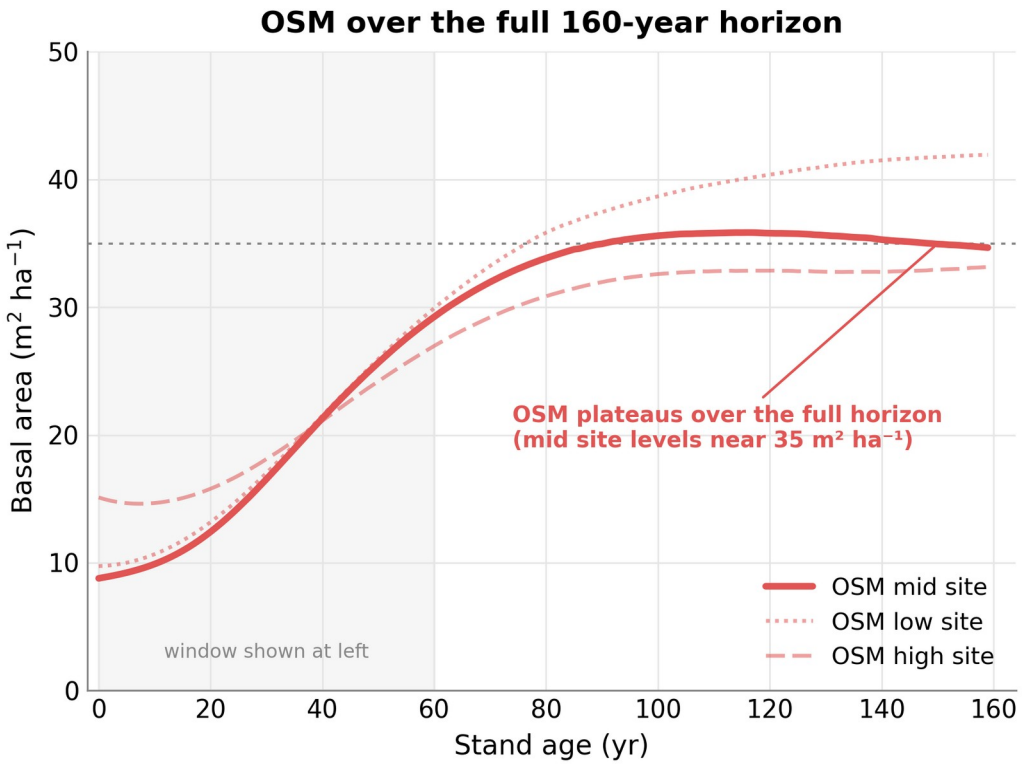
SELF-THINNING TIMING

Long horizon: FVS-ACD self-thins earlier, OSM plateaus later

Self-thinning is a timing difference, not a permanent divergence: FVS-ACD self-thins earlier, OSM plateaus later



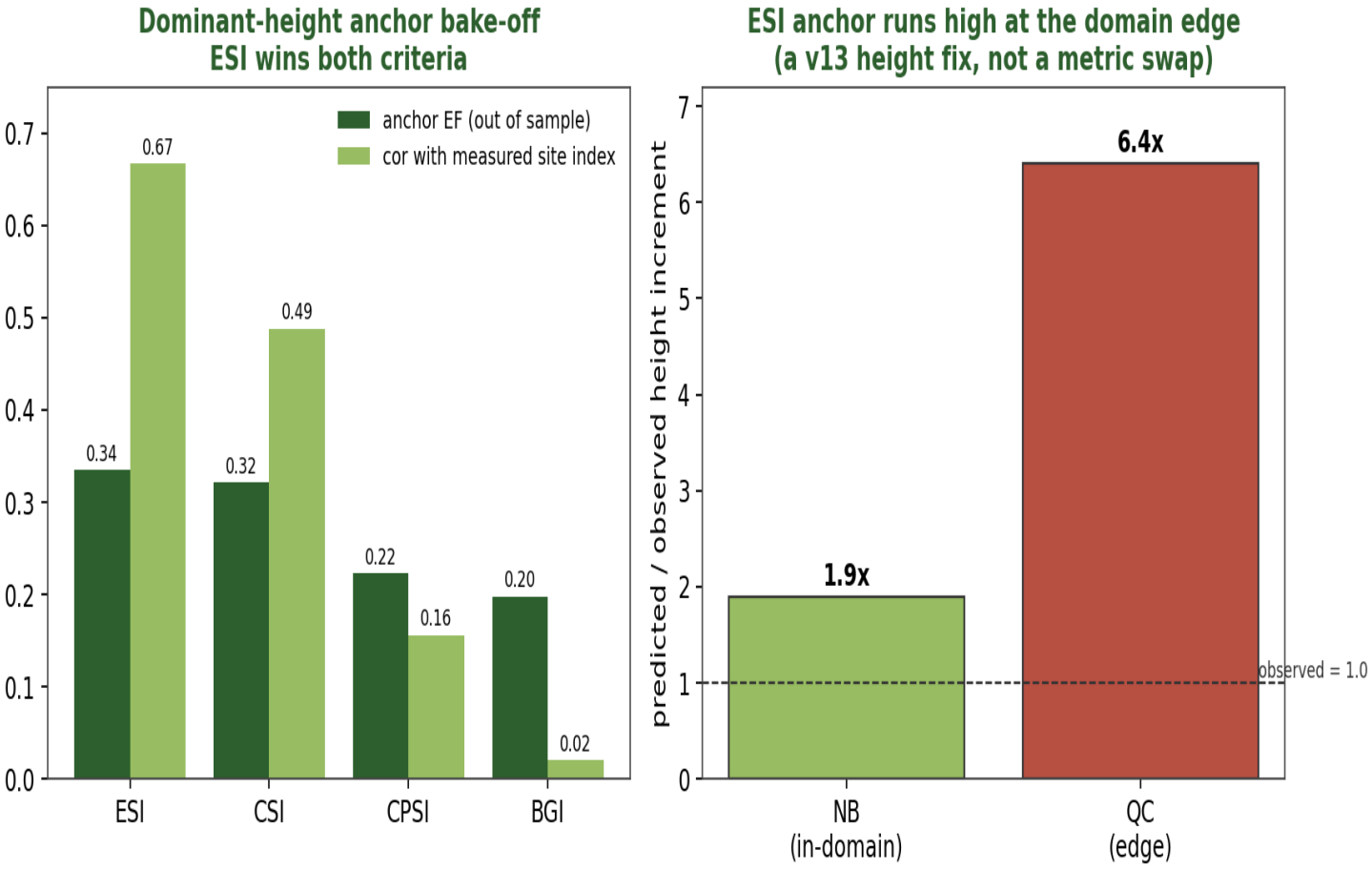
Same nine stands under each model from an identical year-0 start. The age-60 gap is a 35% timing difference.



OSM real-stand productivity terciles (73 benchmark stands). All three level off; the mid site is the panel A analog.

The operational short-interval domain agreement hides a long-horizon divergence in the timing of self-thinning: FVS-ACD self-thins earlier, while OSM plateaus later over the full horizon. This is why FVS-ACD is the choice for carbon, old-growth, and climate work. Scored in volume, the same timing difference makes long-horizon bias flip sign with stand maturity in both models, under-prediction in developing stands and over-prediction in mature reserves.

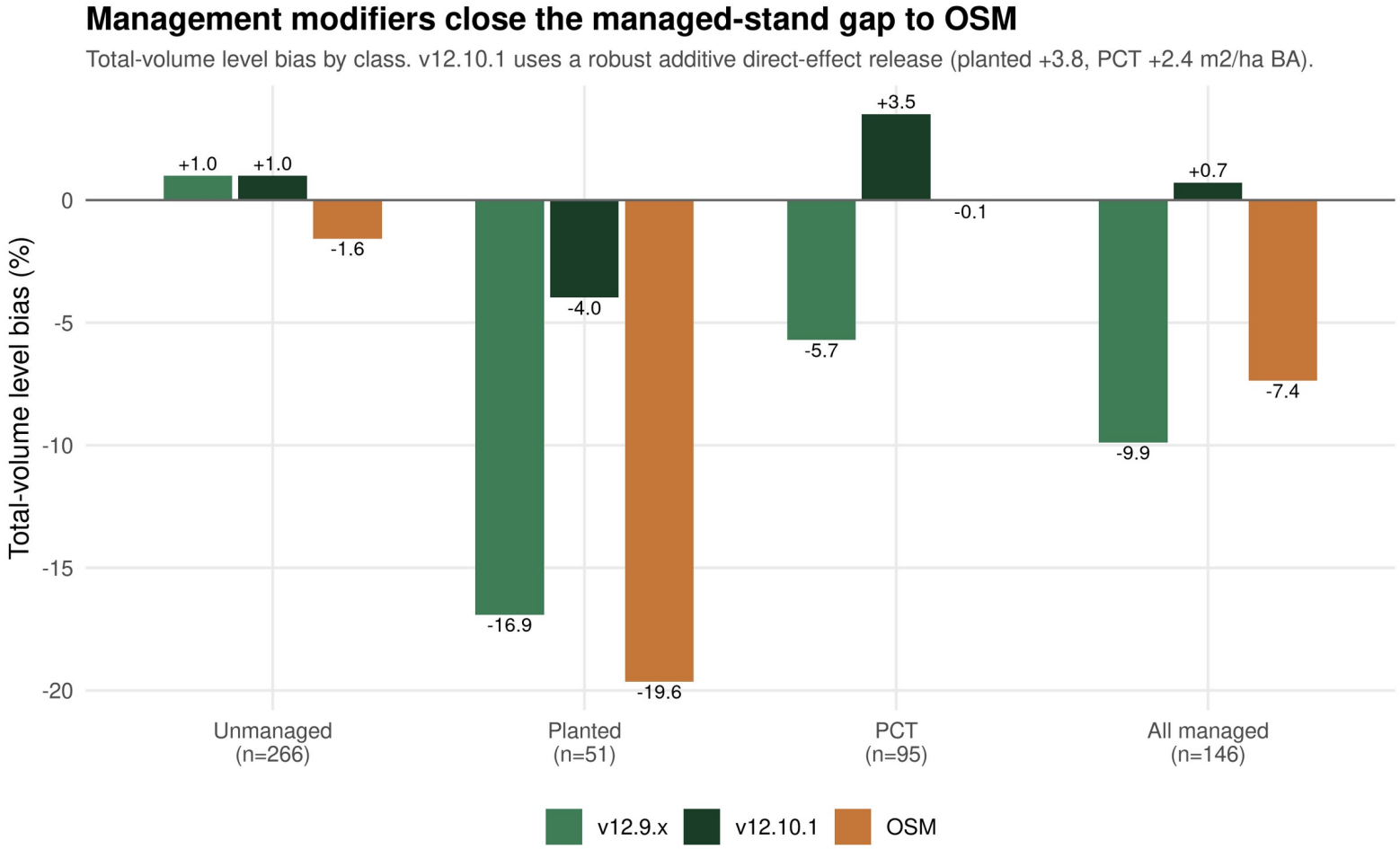
Site productivity: keep ESI



Bake-off settled. ESI vs BGI vs CPSI vs legacy CSI, out of sample with spatially blocked CV. ESI wins the dominant-height anchor and ties for diameter growth. BGI and CPSI carry no height signal and lack cross-border coverage.

Why pooled parity is deferred, not hand-waved. We tested a top-height anchor correction and did not ship it. In the shipped config a +4% top-height over-prediction offsets a −10% basal-area under-prediction, so correcting height alone drives volume further negative. True parity needs a joint height-and-basal-area re-anchor. That is v13.0.

The Chapter-11 lever: a fitted release, not a clamp



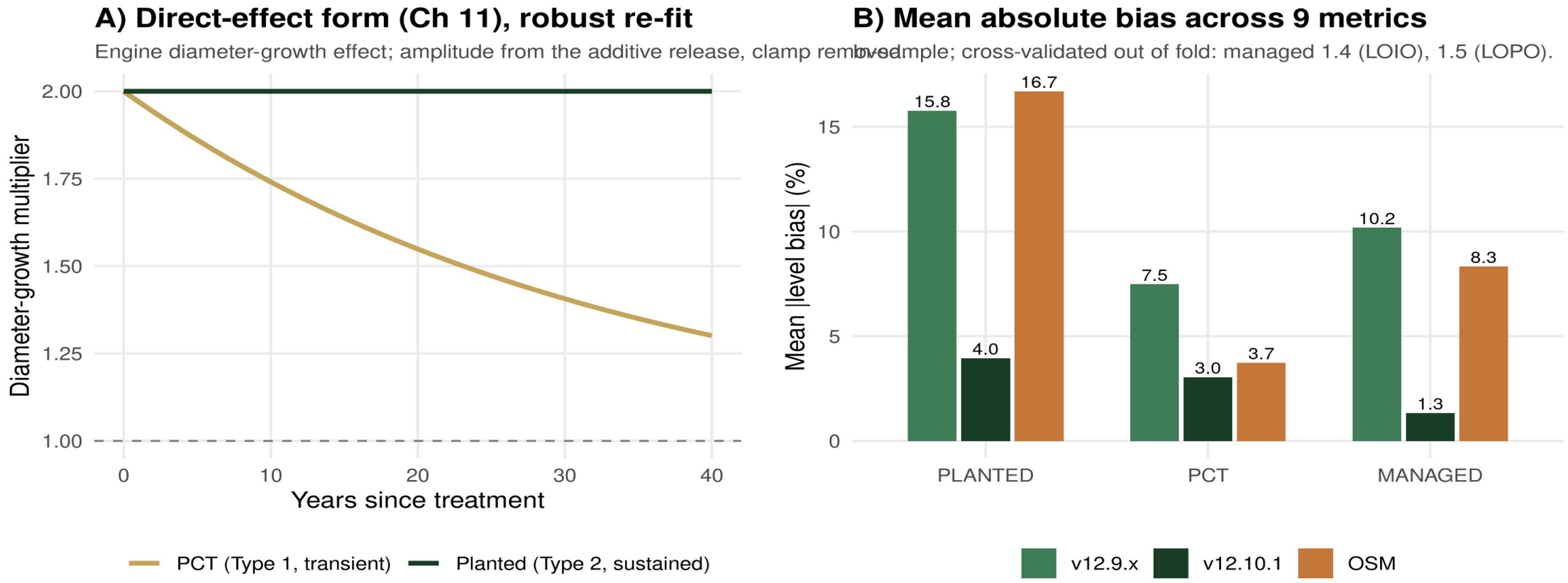
Direct effect only (Ch 11).
The base equations already carry the indirect structural effect. The modifier isolates the direct effect by the multi-step method (Wang 1990).

Planted = Type 2, sustained.
PCT = Type 1, transient, decaying.

Result: robust re-fit. The direct effect is a fitted additive release (planted +3.8, PCT +2.4 m^2 ha^{-1} BA), not a clamped ratio, so its fold coefficients move with the data. v12.10.1 clearly beats OSM on planted and edges OSM on managed out of sample. The effect is larger than the old 1.6 clamp allowed, which is why the capped version still under-predicted planted.

Out of sample it holds: managed mean absolute bias 0.9 installation-out and 1.5 province-out vs OSM's 8.3; planted beats OSM in every scheme (4.0 installation-out, 5.7 province-out, vs 16.7); PCT is 1.5 installation-out, 4.1 province-out (vs OSM's 3.7).

The modifier closes the managed gap



Robust additive direct-effect release (planted +3.8, PCT +2.4 m^2 ha^{-1} BA), clamp removed; engine carries a sustained (planted) or decaying (PCT) diameter-growth effect bounded by self-thinning. In-sample managed mean absolute bias across nine metrics: v12.9.x 10.2 to v12.10.1 1.3 (OSM 8.3). Cross-validated on the next slide. Gated, default off, so unmanaged stands are untouched.

NEW CALIBRATION APPROACHES

Does the lever hold out of sample?

Mean absolute level bias across nine metrics, lower is better. OSM reference is class-specific.

Cross-validation scheme	Planted	PCT	Managed
Leave-one-installation-out	4.0	1.5	0.9
Leave-one-province-out	5.7	4.1	1.5
Managed split-half (200 reps)	n/a	n/a	5.2
OSM reference	**16.7**	**3.7**	**8.3**

What survives the red team

Managed is a clean out-of-sample win: 0.9 leave-one-installation-out and 1.5 leave-one-province-out vs OSM 8.3.

Planted is a robust win over OSM in every scheme (4.0 installation-out, 5.7 province-out, vs 16.7) and transfers across the NB to QC border.

PCT is 1.5 leave-one-installation-out; it now transfers NB to QC at 4.1 leave-one-province-out (vs OSM 3.7), and the PCT basal-area residual closes. Quebec edge still reported as out of domain, conservatively.

The robust additive re-fit removed the 1.6 clamp; fold coefficients now move with the data (planted basal-area release 3.6 to 4.1 across folds) and the out-of-sample numbers hold. In-sample class bias is near zero by construction, so the cross-validated numbers above are the evidence. These planted, PCT, and managed cross-validation results carry unchanged into v12.11: none of the five new levers touches the Chapter-11 direct-effect fit. Figures here are source-verified against the benchmark report (Table 23 and Section 7.9); the split-half figure is the managed scheme only, not run for planted or PCT.

NOVA SCOTIA

The managed-stand levers generalize to a fully independent province

1

4,504 scored pairs, entirely independent of everything else in this report

2,157 managed and 2,347 unmanaged stand-pairs from Nova Scotia's Permanent Sample Plot network, established 1965, the Acadian region's longest continuous forest-inventory record. Never used to fit or tune any v12.11 lever, any base-model coefficient, or the ESI site-index anchor: the cleanest available test of generalization.

2

Managed total-volume bias: -58.10% (base) to -42.32% (v12.11 candidate)

A 27% relative reduction, closely matching the size of the managed-stand improvement seen throughout the rest of this report on data the model was actually fit against. Lever 2's management gate is doing the same job here that it does on Maine, New Brunswick, and Quebec data.

3

Unmanaged bias: -6.04% to -6.07%, essentially untouched

Exactly the behavior Lever 2 is designed to produce: it acts only where a stand is flagged managed and leaves natural stands to the unmanaged self-thinning machinery. Softwood, mixedwood, and hardwood forest groups all improve under the candidate here; the overall NS bias remains large (-41.28% to -30.61%) because it is dominated by the managed subset's larger absolute gap, addressed on the next slide, not a forest-group or hardwood issue.

NOVA SCOTIA

NS site index is measured correctly; most of the gap is real disturbance response, not a site-index artifact

1

Same basis, no species/base-age/unit bug

NS SITE_INDEX and its research-plot proxy AVGTOPHTM are dominant height in meters read at approximately breast-height age 50, the same basis documented for the model's own ESI-anchored dominant-height input. There is no mismatch on the NS side.

2

A transferability caveat: the QC/NB esi8 raster is +7.0 m biased (r=0.146)

The mismatch runs the other way. The QC/NB ESI v8 raster is independently documented (n = 40,596 measured site-index records) to overstate true site index by +7.0 m with essentially no per-plot linear skill. The model's NB_AFAC=0.579 / QC_AFAC=0.111 height-growth anchor exists specifically to counteract that raster's inflation; NS gets no equivalent correction and is fed its real, uninflated field site index directly. Any future region relying on the same pathway inherits this same unproven, poorly constrained correction.

3

The correction is real but secondary, and tested on the unmanaged subset only

Re-scoring the unmanaged NS subset (n = 2,347) with a +7.0 m basis-matching adjustment moves total-volume bias from -6.04% to -4.84% (base) and -6.07% to -5.04% (v12.11), roughly a 15 to 20% relative reduction of an already-small bias. Most of the large -58% managed / -41% overall under-prediction is NOT primarily a site-index artifact: it is dominated by real, short-term post-disturbance growth response. This finding is unmanaged-only; the much larger managed and overall biases were not rescored with corrected site index and should not be over-generalized.

NOVA SCOTIA

Clean long-interval rescore: under-prediction attenuates, it does not flip to over-prediction

1

A disturbance-screening bug was found, fixed, and caught before it shipped

The tree-level cut-detection filter compared a string-typed status column against an integer constant, always false, so it silently passed every stand as undisturbed regardless of harvest history. Correcting it shows 23% of the over-30-year bucket carried an undisclosed harvest inside the interval; an early pass, scored before the fix, appeared to show over-prediction there, an artifact of exactly this contamination, caught in red-team review before it shipped.

2

The clean rescore is complete (jobs 12536574 and 12536575)

The contamination-free pair set (n=5,348: 4,014 unmanaged, 1,334 managed single-regime) was scored through the identical, unmodified frozen harness used everywhere else in this report. Managed single-regime pairs were unaffected by the bug. The over-30-year bucket (n=903) is entirely unmanaged and disturbance-free by construction, mean interval 36.3 years.

3

NS is partially consistent: bias weakens toward zero, it does not flip positive

Total-volume bias at over 30 years is -0.06% (base) and -1.66% (v12.11), essentially zero, not over-prediction; basal area stays under-predicted, -18.08%/-12.48%. Bias attenuates with interval length (-44.04%/-32.38% under 15yr, -56.08%/-48.20% at 15-30yr) but never flips positive, so NS is only partially consistent with the sign-flip pattern seen elsewhere. The managed-lever finding stands: managed bias improves -74.36% to -55.07%, unmanaged stays flat -5.52% to -6.01%.

NOVA SCOTIA

OSM-ACD v2.26.1 head-to-head: a mixed, honest result, not a clean sweep

1

Managed stands: v12.11 and OSM are essentially tied

Short interval: base -58.10%, v12.11 -42.32%, OSM -41.96%, both candidates cutting the frozen base's bias by about 16 points. Long interval: base -74.36%, v12.11 -55.07%, OSM -57.40%, v12.11 modestly ahead. OSM is itself a native Acadian-zone model, so this is a faithful, apples-to-apples comparison, not a proxy.

2

Unmanaged stands: OSM leads at short intervals, v12.11 at long

Short interval: base -6.04%, v12.11 -6.07%, OSM -2.58%, a genuine OSM advantage. Long interval: v12.11 -6.01%, OSM -6.86%, v12.11 slightly ahead. Neither model is uniformly better on unmanaged stands; the direction flips with interval length.

3

Over 30 years: v12.11 holds bias near zero, OSM drifts to -9%

On the disturbance-clean, over-30-year bucket (n=903), Nova Scotia's longest independent record, total-volume bias is -1.66% for v12.11 against -9.14% for OSM, the clearest point of separation anywhere in this Nova Scotia validation. Consistent with the maturity-gated stand constraint (Lever 3b) doing real work at exactly the horizon it targets. Data: figures/_data/ns_headtohead.csv.

INTEGRATING CLIMATE

Climate enters as a scenario driver, not a benchmark

A validated base and a separate scenario driver.

ESI v8 gives current and 2030 climate site surfaces. The projected surface is a forward scenario driver, kept separate from the validated base and never benchmarked on.

For a credible forecast the future step uses a parametric climate-sensitive form (a GAM ratio on the dominant-height anchor), not a random-forest extrapolation. RCP4.5 gives a defensible cold-limited signal of +1.9 m (2030s) to +4.3 m (2090s).

Rule

Validate on observed climate.
Project with a parametric form.
Never benchmark on the projected surface.

AI AND PRODUCTIVITY

AI accelerated the work, humans kept it honest

Accelerated	Weeks of benchmarking, scorecards, and figures compressed into days of iteration.
Kept honest	A red-team pass forced three reframes before v12.11 shipped: top height traced to the ESI anchor, not a lever; Lever 4 rescored as a species-average correction; the Quercus rubra coefficient capped after its field-validation gap surfaced.
Reproducible	Every number traces to a script and a keyed data file on Cardinal; gate-off reproduces v12.10.1 bit for bit.

ROADMAP

v13 is the fall arc: what the data lets us commit to now

1	**v13.0 Height and domain**	Bound the calibrated domain: gate off predictions outside the ESI range so the anchor never extrapolates. The height re-anchor blend sweep is exhausted and parked: every blend traded a top-height gain for a volume loss, so none ships. The basal-area correction is retained and the domain gate stands. Mature Acadian data still identifies the asymptote for the New Brunswick to Quebec transfer.
2	**v13.1 Hierarchical mortality and ingrowth**	Gated and running (job 12470707). Targets what v12.11 did not close: the Quebec management-gate volume over-correction, SILC's basal-area widening, and the developing-end gap at both whole-plot and subplot grain. Carries a region random slope on management (SD 1.21 on the log-hazard scale, far larger than the mean effect), a cohort exemption for Lever 3b's reallocation pool, and recalibrated Russell, Weiskittel and Kershaw (2014) crown-ratio recession coefficients.
3	**The ask of the group**	Provincial managed remeasurements and PSP treatment histories, with the provincial site surfaces. This is the data that unblocks the New Brunswick to Quebec management transfer, the one place the managed calibration still leans on thin planted and PCT samples. August is the moment to request a harmonized data-sharing arrangement so v13 is fit on the full Acadian record.

VERDICT

v12.11 is ready: it corrects the bias it targeted, and says so honestly

- Every lever is gated and off by default; gate-off reproduces the frozen v12.10.1 baseline bit for bit (0.000e+00 max absolute difference), reconfirmed on a novel stand after the red-team fixes. All four guardrails pass: frozen in-domain FIA NSVB, the Bakuzis five relations, the long-horizon plateau, and managed out-of-sample performance against OSM v2.26.1.
- On the sites it was built for, managed CTRN and NB MagPlot, and the held-out mature Baxter reserve, v12.11 roughly halves bias against the frozen base and now beats OSM rather than trailing it. Two managed flags, Quebec's volume over-correction and SILC's basal-area widening, are disclosed and named as v13.1 targets, not hidden.
- On developing Maine FIA, OSM still leads, and moving to the finer subplot grain does not close the gap (-24.90% to -24.82%, a 0.08 point change). This is the acknowledged open gap in this candidate, and the explicit target of v13.1, not a claim closed by v12.11.
- v13.1's hierarchical mortality and ingrowth refit is already gated and running. It is the fall arc: it targets the Quebec and SILC flags and the developing-end gap directly, then management enters as a growth-scale release in v13.2, with prediction intervals and climate following.
- This edition adds Nova Scotia, the region's longest CFI record: the managed-stand levers generalize cleanly, site-index is answered honestly, and the long-interval test, cleaned of a contamination artifact, is only partially consistent with the pattern seen elsewhere: bias weakens, it does not flip positive.

Questions / Comments

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All results reproducible and openly archived; plot coordinates stay on the analysis cluster, aggregates only.

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